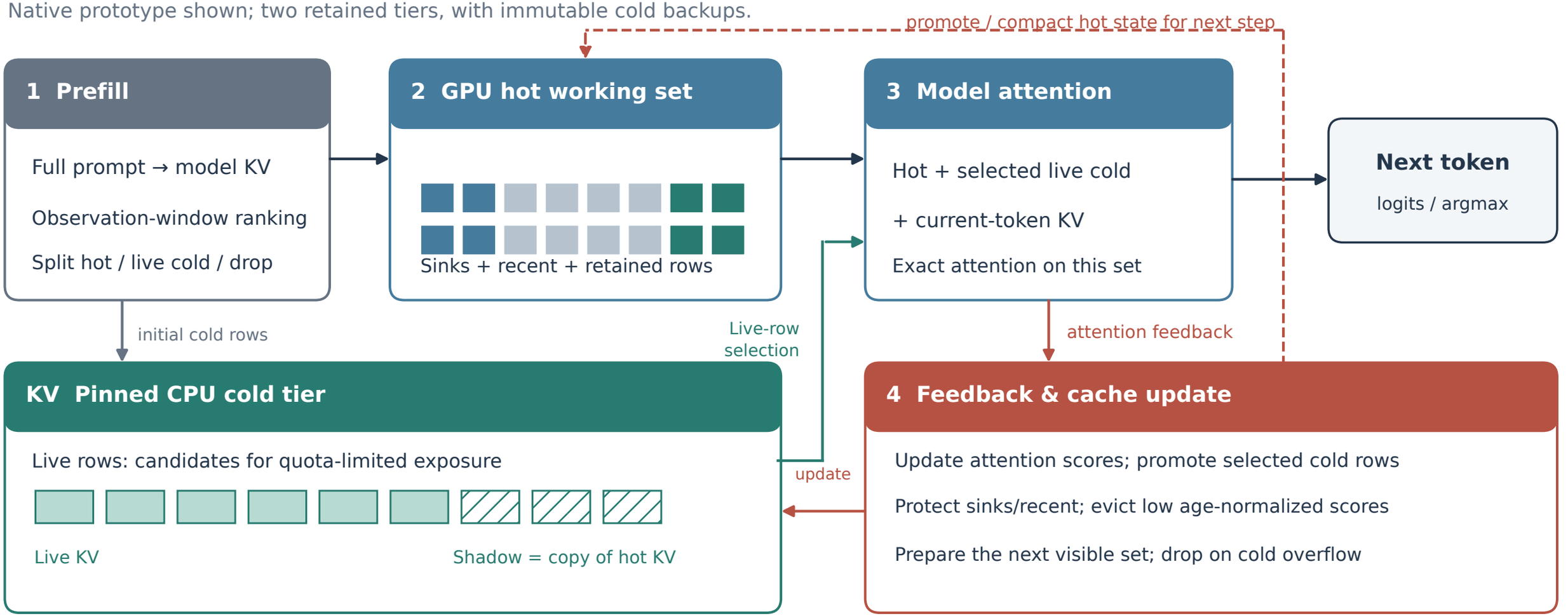
TieredKV: data movement and write-aware cache control

Native prototype shown; two retained tiers, with immutable cold backups.



Hot victim already has a shadow? YES Reactivate; no KV payload write NO Append cold KV

No third storage tier: excess cold rows are permanently dropped.

STT-RAM latency/energy are analytical assumptions; this implementation uses CPU DRAM. Default functional mode stores cold KV on GPU.